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Improving Indoor Fingerprinting Positioning With Affinity Propagation Clustering and Weighted Centroid Fingerprint

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ABSTRACT Nowadays, research and development of various indoor positioning systems (IPS) have been increasing owing to flourishing social and commercial interest in location-based services (LBSs). Among LBS technologies, we used the Bluetooth low energy beacon in our system, which consumes less energy and is embedded in many current smartphones and tablets. In particular, the fingerprinting method has become a prime choice in the design of IPS owing to its good location estimation and the fact that a line-of-sight from access points is not required. We propose an improved two-step fingerprinting localization using multiple fingerprint features to enhance the localization accuracy. The proposed system uses a propagation model to convert RSS of beacons to distance and estimate the weighted centroid (WC) of nearby beacons. The estimated WCs along with signal strength and rank of the nearby beacons are stored in the server database for localization instead of RSS from all the deployed beacons. First, the proposed system makes use of diverse fingerprinting features to increase localization accuracy that also reduces both the physical size of the database and the amount of data communication with the server in the execution phase; second, affinity propagation clustering minimizes the searching space of RPs and reduces the computational cost; third, exponential averaging is introduced to smooth the noisy RSS. The experimental results obtained by real field deployment show that the proposed method significantly improves the performance of the positioning system in both the positioning accuracy and radio-map database size.

INDEX TERMS Affinity propagation clustering, BLE, Exponential averaging, RSS, Weighted centroid.

I. INTRODUCTION

A location-based-service (LBS) system obtains a mobile device location and provides it to a user. The location information at outdoor and indoor is the core of ergonomic future sustainability computing framework, which provides rich possibilities for a pervasive computing environment [1]. Meanwhile, the global navigation satellite system (GNSS) has been a dominant technology in an outdoor localization system. However, the inapplicability of GNSS in indoors invites an extensive study and the development of a distinct indoor positioning system (IPS).

Diverse types of technologies and methods are adopted for the development and implementation of IPS such as

Wi-Fi, Bluetooth Low Energy (BLE), ZigBee, ultra-wide band (UWB), visible light wave, and geomagnetism [2]–[7]. The study of IPS can be divided into probabilistic and deterministic works [8]. The former yields localization information by estimating a probability distribution over the reference points, whereas the latter implements certain distance conversion or fingerprinting data comparison algorithms to find an estimated location [9]. The deterministic method of IPS is further categorized as propagation model and fingerprint-based methods.

Trilateration and triangulation need a proper distance estimation process between a tag and anchor devices based on a signal propagation model. Similarly, fingerprinting involves the comparison of user data with those in the database called radio map. Fingerprinting is the pervasive method governing IPS owing to its high reliability. However, it also comes with

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a time consuming and labor intensive offline/training phase as a tradeoff. The data employed in fingerprinting are radio signal strength (RSS) [10], [11], channel state information (CSI) [12], and earth's magnetic field strength [13], [14].

In the conventional approach for fingerprinting, RSS from all the deployed access points (APs) is accumulated and stored in the database during the offline/training phase. The database contains coordinates of reference points (RPs) and the average RSS from each AP that is measured individually at all RPs. During the online/execution phase, the observed RSS is compared against the stored one in the database, and then the coordinate with the closest match is considered as the estimated user location. In general, weighted k -nearest neighbor (Wk-NN) algorithm is used in traditional fingerprinting where k RPs are selected based on similarity of their fingerprints with the measured RSS. The location is then estimated as a weighted sum of the real locations of the k RPs. This approach of Wk-NN is also called flat Wk-NN where all the RPs on the testbed are considered for final location estimation. The flat Wk-NN can be improvised into two-step Wk-NN by employing a clustering method where RPs are divided into clusters. In two-step Wk-NN, only cluster RPs are considered for positioning estimation that leads to significant computational complexity reduction and improved localization accuracy compared to flat Wk-NN. Moreover, in the conventional Wk-NN, only a single feature (e.g. average RSS) is used to define the fingerprint of an RP.

In this paper, in addition to the average RSS, we propose to use rank and weighted centroid (WC) fingerprint in IPS accompanied by affinity propagation clustering (APC) for better positioning accuracy. In our system, instead of storing RSS from all the APs, we use the RSS data to divide the whole testbed into several clusters employing the APC. Then, RSS from u beacons ($u < \text{total number of deployed beacons}, B$) that are close to real location of the tag device is converted to a distance using a propagation model. Later, the distances from APs to the tag device are converted to their respective weight to find the WC of nearby u beacons [2]. The WC is estimated using the known APs (beacons) locations and their respective weights at each RP. Hence, the fingerprinting data for an RP contains the estimated WC, average RSS, and rank of nearby u beacons along with real location of the RP. Note that the rank of beacons refers to sorted beacons by their descending RSS value where first u beacons in the sorted list are selected as nearby u beacons. During the testing phase, cluster-head is first searched employing the online WC. Second, rank of online seen u beacons are compared with the stored ones and a match ratio is obtained. The match ratio is further employed to assign a proportional weightage to WC-based and RSS-based fingerprinting to yield the final location estimation. We claim the following contributions in this paper.

- 1) We use diverse features (WC, average RSS, and rank) to define a fingerprint of an RP to yield better positioning result compared to the existing methods. We employ APC as a clustering component where WC

estimation is used as cluster head determination metric on the online phase. The fluctuating RSS samples are smoothed with Exponential averaging (EA) filter.

- 2) The proposed method is realized employing only the signals from nearby u beacons that are at the vicinity to the tag device. On the training phase, as compared to conventional Wk-NN, the signals from $B - u$ beacons is redundant to the proposed method. It helps to significantly reduce the physical size of fingerprint database. On the other hand, attenuated signals coming from distant beacons plays no role on localization estimation on execution phase. It results in better localization estimation.
- 3) The results are based on real world deployment with modern iOS phone.

The remainder of this paper is organized as follows. Section II briefly describes some published literature on indoor fingerprinting localization. The background behind WC and APC is elaborated in Section III. Section IV describes the proposed positioning method using WC fingerprint. Experimental results and discussion are presented in Section V, and finally, the conclusion of the paper is presented in Section VI.

II. RELATED WORK

Generally, a fingerprinting IPS consists of two phases of operation, namely offline/training and online/execution phases. A conventional flat Wk-NN is presented by Pavel *et al.* using BLE beacons in [15]. Their method sorts the database fingerprint according to the Euclidean distance to online RSSI. Among them, the first k fingerprints are selected for a weighted estimate of the position. Furthermore, they combined the BLE with existing Wi-Fi to obtain improved localization estimation. Another study on beacon-based fingerprinting used the probabilistic approach for localization estimation [16]. Here, a Gaussian kernel was used as a basis function in the Bayesian likelihood function to generate a probability for each RP. An approach in IPS utilizing the beacon's location and estimated distance between the beacon and the tag device is put forward in [17]. This work converts the acquired RSS into distance employing propagation model and sends the estimated distance along with the beacon's location coordinate to the server through a cellular or Wi-Fi network. The received data at the server is further fed to extended kalman filter to produce location information. Another example employing true position coordinate of beacons for localization estimation is [18]. Here, pedestrian dead reckoning (PDR) is used with BLE beacons where the proximity information of beacons is used for correction of estimated position by PDR.

Conventional Wk-NN fingerprinting has also been realized with Wi-Fi signals. Recent advances on Wi-Fi fingerprinting localization is elaborated in [19]. This survey work overviews on advanced localization techniques and efficient system development using the Wi-Fi signals. References [20] and [21] present improvisation methods over

the conventional Wk-NN fingerprinting using Wi-Fi signals. Reference [20] uses average RSS and standard deviation of Wi-Fi signals at RPs from APs to form a fingerprint. On online phase, both the average RSS and standard deviation is processed to estimate a Euclidian distance. Based on the estimated Euclidean distance, k RPs are selected to estimate a coarse position. Furthermore, a joint probability for each RP is calculated, k RPs are selected based on their joint probability to estimate another coarse position. Both the coarse localization estimations are fused together using shortest Euclidean distance and largest joint probability to yield a final localization estimation. Meanwhile, [21] proposes to use Manhattan distance instead of Euclidean distance to compare the closeness of online received Wi-Fi signals strength with the stored fingerprint.

RSS clustering is used to improve the indoor fingerprinting positioning [22]. A typical method of RSS clustering is to choose a set of cluster centers to reduce the sum of squared distances between the RSS data and their corresponding centers. For instance, K-means clustering begins by choosing both the number of output clusters and the corresponding set of initial exemplars, where it iteratively refines the output clusters to decrease the sum of the squared distances [23]. The K-means clustering is sensitive towards the initial selection where the clustering operation is performed several times with different initial centers to yield the best clustering result [24]. The requirement of an arbitrary selection of initial cluster centers adds a limitation to the K-means clustering. However, APC starts by assigning each RP the same chance to become an exemplar (cluster center). The APC creates the centers and the corresponding clusters depending on message exchange of RSS readings similarities between the RPs. In addition to APC and K-means, fuzzy c-means and hierarchical clustering strategy (HCS) are some representative methods of clustering [25]–[28]. Reference [29] utilizes K-means and fuzzy c-means clustering in wireless sensor networks (WSN) to divide the fingerprinting database into clusters. For each cluster, an artificial neural network is trained by using only those fingerprints that belong to the cluster. The final location is estimated by comparing the online signal strength measurements with the prototype signal strength vector that was determined for each cluster. HCS has been put forward in [28] where it groups the training locations (RPs) into a cluster if those training locations detect strongest RSS from one particular AP. The HCS starts with assigning total number of APs as the number of clusters and later it multiplies to more clusters depending on second highest RSS from neighboring APs.

APC is employed in [24], [30], and [31] for clustering the testbed using Wi-Fi RSS data. All of this literature estimate the cluster-head on online phase based on nearest RSS fingerprint to the online-acquired RSS data. Reference [30] used APC with Wk-NN algorithm which is the deterministic-based whereas [24] and [31] employed APC with probabilistic-based fingerprinting.

III. BACKGROUND

A. WEIGHTED CENTROID IN FINGERPRINTING POSITIONING

WC estimation gives an approximation of a shifted centroid value among u anchors, where a weight in WC is determined for each anchor at the tag location [2]. The simplest WC estimation is defined by the following sets of equations:

$$\begin{aligned} x_w &= \frac{\sum_{j=1}^u x_j \times w_j}{\sum_{j=1}^u w_j} \\ y_w &= \frac{\sum_{j=1}^u y_j \times w_j}{\sum_{j=1}^u w_j} \\ w_j &= \frac{1}{d_j^g} \end{aligned} \quad (1)$$

where (x_w, y_w) is the estimated WC, (x_j, y_j) is the previously known anchor coordinate, d_j is the distance between the tag device and anchor (beacon) j , g is the degree of weight, and u is the total number of beacons considered for WC estimation.

Fig. 1 illustrates the WC estimation mechanism. WC has been used as a localization estimation method in existing studies with many wireless technologies such as wireless sensor networks, ZigBee, and BLE [4], [32], [33]. Essentially, WC estimation has the following characteristics:

- 1) The estimated location is confined inside the beacon locations only.
- 2) The estimated location is dragged toward the nearest anchor owing to its large weight.

Moreover, the degree of weight (g) can be adjusted as per the distance between the deployed beacons. A large g value drives the WC estimation very close to the location of the nearest beacon, whereas a very low value (close to zero) yields a geometrical centroid. Typical values of g are 0.5, 1, and 2.6 [2], [32], [34], [35].

The features defining the RPs of a testbed should have two basic characteristics namely: spatial diversity and temporal stability [36]. The WC information on the testbed can be approximated as spatially diverse and temporally stable. Since WC consists of 2D Cartesian coordinates, it has the benefit of two degrees of freedom. Moreover, when two different sets of beacons are seen in two regions inside a testbed, they cannot have the same set of WC. In contrast, for the same set of beacons seen in a testbed region, we can infer from (1) that (x_j, y_j) is fixed for any two adjacent RPs. Hence, their WC depends only on w_j . In addition, the weight (w_j) of a beacon is inversely proportional to the distance between the beacon and the tag device and the distance (d_j) is derived from RSS.

B. AFFINITY PROPAGATION CLUSTERING

In general, APC splits up a set of elements into clusters and chooses a representative cluster-head for each cluster [25]. The APC starts by allocating every element the same opportunity to become a cluster-head [23]. Let us assume that the

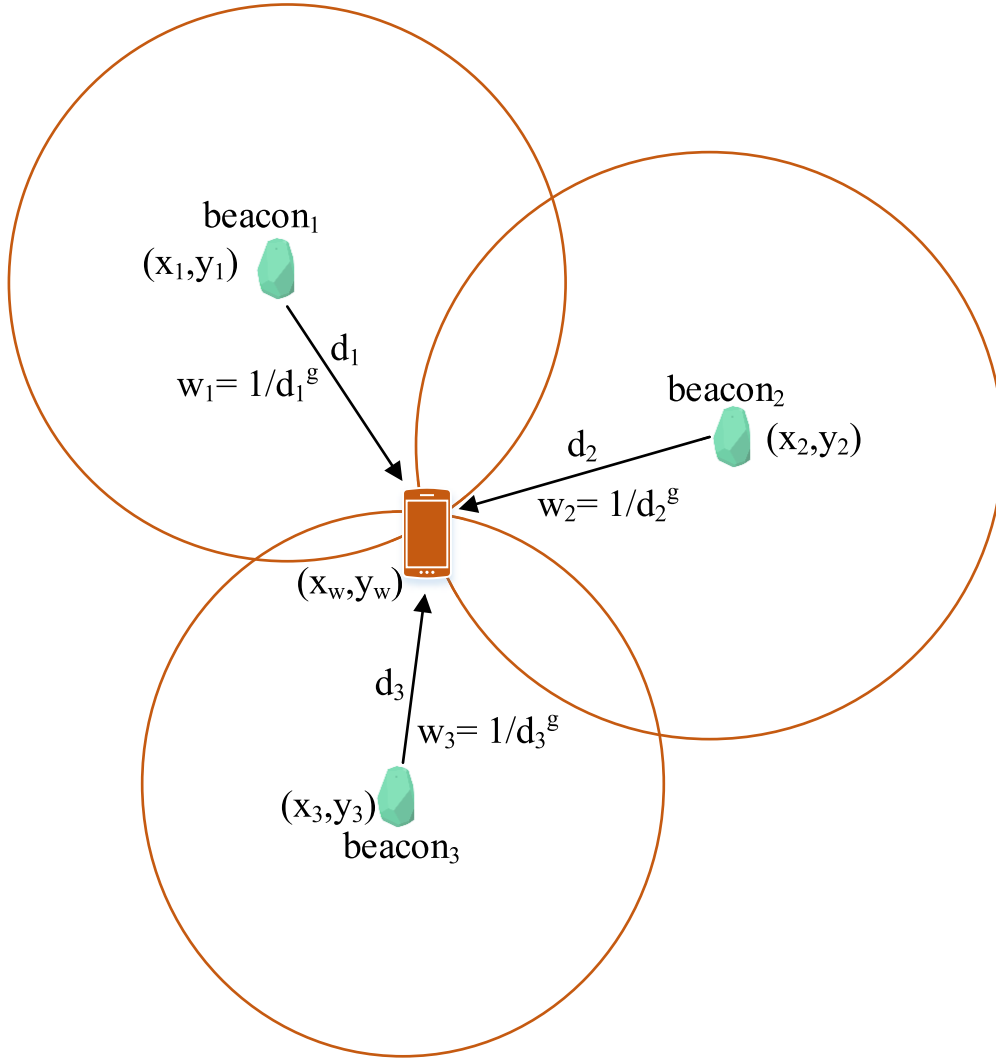


FIGURE 1. Procedure for estimating weighted centroid at a tag device.

radio map formed by RSS acquisition on the training phase is represented by following relation:

$$S_{RSS} = \begin{pmatrix} S_{1,1} & S_{1,2} & \dots & S_{1,N} \\ S_{2,1} & S_{2,2} & \dots & S_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ S_{B,1} & S_{B,2} & \dots & S_{B,N} \end{pmatrix} \quad (2)$$

where, $S_{b,j}$ ($b = 1, 2, \dots, B; j = 1, 2, \dots, N$) is the average RSS obtained from beacon b at the j^{th} RP (RP_j).

The fitness of a RP_j to be selected as the cluster-head with respect to RP_i is defined by squared Euclidean distance known as pairwise similarity $sim(RP_i, RP_j)$ (where $i \neq j$) given as:

$$sim(RP_i, RP_j) = -\|\vec{S}_i - \vec{S}_j\|^2, \quad \forall i, j \in \{1, 2, \dots, N\} \quad (3)$$

where $\vec{S}_j = [S_{1,j}, S_{2,j}, \dots, S_{B,j}]^T$.

Moreover, the preference (*Pref*) or self-similarity can be defined as:

$$Pref = median\{sim(RP_i, RP_j), \quad \forall i, j \in \{1, 2, \dots, N\}, \quad i \neq j\} \quad (4)$$

The APC transmits two different kinds of messages among the RPs during its operation. They are responsibility message ($r(RP_i, RP_j)$) and availability message ($a(RP_i, RP_j)$), and are iteratively updated by the following relations:

$$r(RP_i, RP_j) = sim(RP_i, RP_j) - \max_{j' \neq j} \left\{ a(RP_i, RP_{j'}) + sim(RP_i, RP_{j'}) \right\} \quad (5)$$

$$a(RP_i, RP_j) = \min \left\{ 0, r(RP_j, RP_i) + \sum_{i' \neq i, j} \max \{ 0, r(RP_{i'}, RP_j) \} \right\} \quad (6)$$

Here, the responsibility message contains the information about the clustering center whereas the availability message informs the attachment relations between the RPs and clusters. Furthermore, a damping factor $\gamma \in [0.5, 1)$ is used to avoid the possible ringing oscillations during updating the responsibility and availability messages. Finally, after introduction of γ , the two messages become as follows:

$$r_t(RP_i, RP_j) = \gamma * r_{t-1}(RP_i, RP_j) + (1 - \gamma) * r_t(RP_i, RP_j) \quad (7)$$

$$a_t(RP_i, RP_j) = \gamma * a_{t-1}(RP_i, RP_j) + (1 - \gamma) * a_t(RP_i, RP_j). \quad (8)$$

Generally, APC connects all the points in the large space and makes each node (RP in our case) a potential exemplar. RPs launch responsibility (5) and receive availability (6) constantly that continue to extend the gap between the exemplar and subsidiary RPs until the exemplar RP is determined finally. For example, the exemplar is updated according to the value of $r(RP_i, RP_j) + a(RP_i, RP_j)$. Hence for RP_i , if $r(RP_i, RP_j) + a(RP_i, RP_j)$ is the largest, it shows that RP_j is the exemplar of RP_i .

IV. THE PROPOSED POSITIONING METHOD

The proposed method has two phases: training/offline and execution/online. In both phases, a propagation model is used for the conversion of scanned RSS values to distance. The propagation model used [17], [37], [38] is described as follows:

$$P_r(d) = A - 10 \times n \times \log_{10}\left(\frac{d}{d_0}\right) + \chi_\sigma, \quad (9)$$

where $P_r(d)$ is the RSS in dBm at distance d , A is the received RSS at a standard distance (defined as 1 m), χ_σ is a Gaussian-distribution random variable with mean 0 and variance σ^2 , and n is the path loss exponent. From 9, we can derive:

$$d = d_0 \times 10^{\frac{A - P_r(d)}{10 \times n}} \quad (10)$$

The measured distance (d) is used to obtain the weight (w_j) in (1). With the previously known locations of APs and their respective weights, the WCs are estimated as given in (1). Note that APs locations are acquired only once on the offline phase and inscribed in the smart phone application, and the proposed method performs with received RSS data on the execution phase.

Fig. 2 illustrates the positioning estimation procedure of the proposed system. In the training phase, at first, the testbed is divided into equal sized N imaginary grids where the center of each grid represents an RP. Therefore, the testbed contains N RPs in locations $L_r(r \in \{1, 2, \dots, N\})$ and the RSS data of B beacons are acquired at r^{th} RP q times; that is:

$$AP_r = \{ap_{r,1}, ble_{r,2}, \dots, ap_{r,B}\},$$

where

$$ap_{r,i} = \text{Avg}(ap_{r,i}^1, ap_{r,i}^2, \dots, ap_{r,i}^q)$$

Here, RSS data of B beacons are used for the APC and only u beacons that are selected based on largest RSS data ($u < B$) are used for fingerprinting information. Note that the RSS data is acquired from four directions at an RP and the tag device is kept in 'texting/messaging' position.

The observed online RSS data at the unknown location r' inside the testbed can be represented as:

$$AP_{r'} = \{ap_{r',1}, ap_{r',2}, \dots, ap_{r',B}\},$$

where

$$ap_{r',i} = EA(ap_{r',i}^1, ap_{r',i}^2, \dots, ap_{r',i}^{t-1}, ap_{r',i}^t).$$

Here t is time instant and EA is the exponential averaging which can be termed as formalization of the simple learning process since it yields the smoothed RSS as soon as two observations are available [39]. The EA for a RSS data sample at an instant $t(RSS_{EA_t})$ is given by the following formula:

$$RSS_{EA_t} = \beta \times RSS_t + (1 - \beta) \times RSS_{EA_{t-1}}, \quad (11)$$

where β is the smoothing factor ($0 < \beta < 1$). Note that a larger value of β reduces the level of smoothing, whereas a value of β close to zero has greater smoothing effect and is less responsive to recent RSS observations.

Let us define the set of cluster-heads as H and the number of RPs that are members of the cluster head as N' ($N' < N$). At first, the cluster-head (RP_H) is determined with the following way:

$$RP_H = \text{argmin}_{m \in H} \|WC_{r'} - WC_m\|, \quad (12)$$

where $WC_{r'}$ and WC_m denote online observed weighted centroid from nearby u' beacons at unknown place r' and the stored weighted centroid of m^{th} cluster-head RP, respectively. Note that u and u' are seen beacons of same size at training and testing phase, respectively.

After the cluster-head RP is chosen, Equidistant distances are calculated depending on online-observed fingerprint data and stored fingerprinting data for N' RPs as shown below:

$$D_{WC_l} = \|WC_{r'} - WC_p\|, \quad p \in N' \text{ and } l = (1, 2, 3, \dots, N') \quad (13)$$

$$D_{RSS_l} = \sum_{b=1}^u \sqrt{(ap_{b,r'} - ap_{b,p})^2}, \quad p \in N' \text{ and } l = (1, 2, 3, \dots, N') \quad (14)$$

where p is a member of the selected cluster head.

The N' RPs are arranged with ascending order of D_{WC_l} and the first k RPs with their known positions $J_z[x_z, y_z]$ are selected to estimate T_{WC} using the following relation:

$$T_{WC} = \frac{\sum_{z=1}^k J_z G_{WC_z}}{\sum_{z=1}^k G_{WC_z}}, \quad \text{where } G_{WC_z} = \frac{1}{D_{WC_z}}. \quad (15)$$

Similar operation is repeated using D_{RSS_l} (u in (14) is the stored beacon at the RP for least D_{WC_l}) to estimate T_{RSS} using

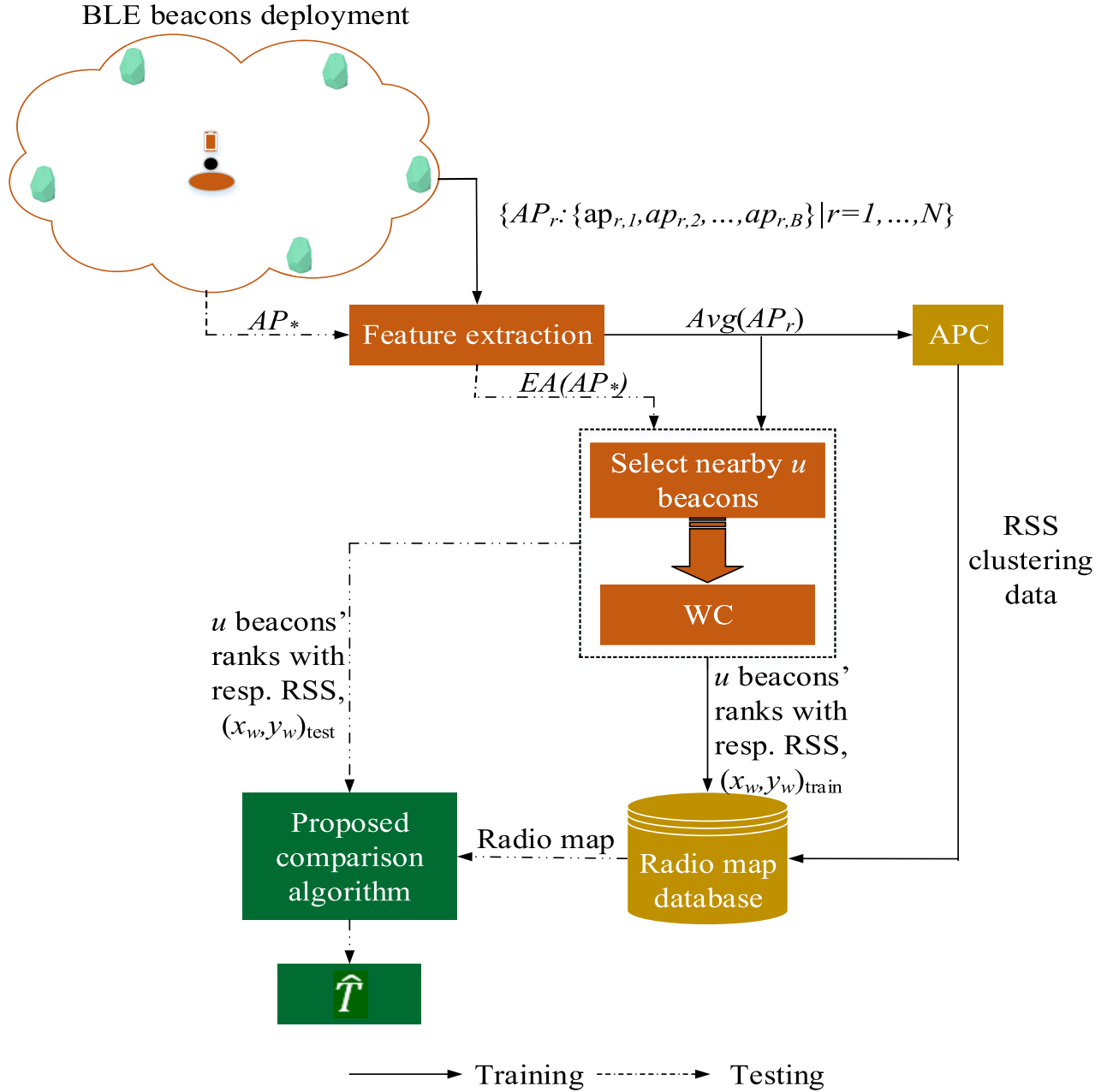


FIGURE 2. Process flow of the proposed fingerprinting IPS.

the following relation:

$$T_{RSS} = \frac{\sum_{z=1}^k J_z G_{RSS_z}}{\sum_{z=1}^k G_{RSS_z}}, \quad \text{where } G_{RSS_z} = \frac{1}{D_{RSS_z}}. \quad (16)$$

The final location is estimated using the following relation:

$$\hat{T} = (1 - \alpha) \times T_{RSS} + \alpha \times T_{WC}, \quad (17)$$

where $\alpha = \frac{u''}{u}$. Here, u'' represents the total number of matched beacons, i.e., number of online-observed beacons at location r' that matches its rank to the stored ones for the RP with least D_{WC_l} . For example, let the online seen

beacons ($u = 4$) at unknown location r' arranged in descending order of their ranks are $\{ap_{r',1}, ap_{r',5}, ap_{r',3}, ap_{r',8}\}$, and the stored rank of beacons at the RP for least D_{WC_l} are $\{ap_{r,5}, ap_{r,1}, ap_{r,3}, ap_{r,4}\}$. Here, since only one beacon matches between online and offline ordered rank (the third beacon), the value for u'' is one.

V. EXPERIMENTAL ANALYSIS

A. EXPERIMENTAL SETUP

The proposed positioning system was evaluated on the eighth floor of IT building, Chosun University, South Korea.



FIGURE 3. Testbeds where the yellow dots represent the deployed BLE beacons. (a) Left from the center of the corridor. (b) Right from the center of the corridor. (c) Panoramic view of a fully furnished computer lab.

Fig. 3 shows the testbed representing a research lab and the hallways.

The BLE beacons were deployed in the rectangular fashion on the ceiling at 2.7 m height. Moreover, the advertisement interval of our beacons was set as 300 ms (three beacon packets per second), and our tag device (iPhone in this work) scans beacons with a 1 second interval on average. The distance between two adjacent beacons was 6.8 m. The whole testbed was divided into grids of size 1.35 m length and breadth where each grid represented an RP.

1) RSS CLUSTERING USING APC

After acquiring the RSS data at all the RPs from all the deployed beacons, RSS clustering operations was performed using APC. As the damping factor is put forward while updating the messages in APC, we analyzed the APC operation by changing the damping factor ($\gamma \in [0.5, 0.9]$) to obtain the minimum number of iterations as shown in Fig. 4.

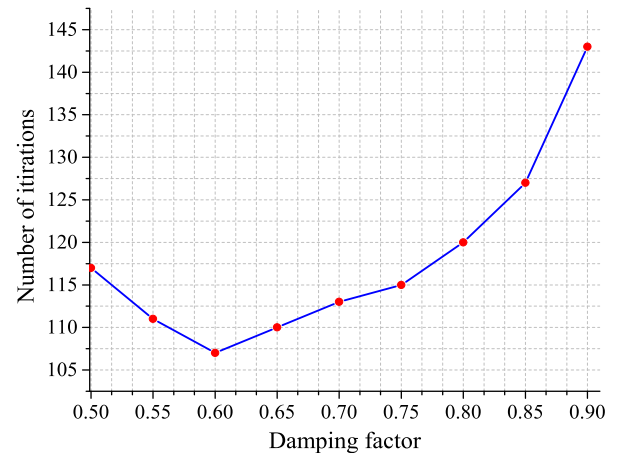


FIGURE 4. Variation on number of iterations with respect to the damping factor.

As shown in Fig. 4, the damping factor with $\gamma = 0.6$ performs the RSS clustering with least number of iterations or least computational cost. Hence, $\gamma = 0.6$ was used for APC operation in this work.

In the conventional fingerprinting or flat-based fingerprinting localization, the computational complexity grows with the size of the radio map (number of RPs on the radio map) [3], [22]. For N RPs on the testbed, the computational complexity of traditional fingerprinting localization (linear matching process) is $O(N)$ [40], [41]. However, the complexity of localization estimation process based on cluster analysis (every cluster having the same number of RPs) is $O(N/H)$, where H is the total number of exemplars. In the proposed method, the RPs under the cluster head N' ($N' \ll N$) can vary and has a computational complexity of $O(N')$. In other words, the coarse localization stage of the proposed method reduces the area of interest from N RPs to N' RPs, which reduces the number of fingerprints to be compared for fine localization and eventually reduces the computational complexity of the system.

2) RSS SMOOTHING

In general, RSS fluctuates over time even in fixed indoor testbeds, resulting in increasing localization estimation error. The reasons that RSS exhibits such high variability in space and time include various noise factors, interferences, and attenuation. In terms of distance estimation using the measured RSS, an RSS smoothing process is required.

We compared the performance of moving average (MA), weighted moving average (WMA), and EA filters.

TABLE 1. Weighting factors and window sizes for the experiment.

Window size	Set of weighting factors
4	[0.4, 0.3, 0.2, 0.1]
5	[0.3, 0.2, 0.2, 0.2, 0.1]

The window sizes and their respective weighting factors for MA and WMA are listed in Table 1. We set $\beta = 0.5$ in EA smoothing to give equal preference to the latest RSS observation and previous RSS smoothed result.

We measured 100 samples of smoothed RSS values using MA, WMA, and EA smoothing filters at the hallways and the computer lab. Here, the tag device was held at a fixed distance to the AP. We calculated the standard deviation (SD) of these samples as a measure of the smoothness of the filter. The observed values are presented in Table 2. Here, (W)MA-4 and (W)MA-5 denote the (W)MA filters with window sizes of 4 and 5, respectively.

TABLE 2. Standard deviations of RSS samples from various filters in the corridor.

Filters	Computer lab	Corridor
Raw RSS	3.025214	1.710898
MA-4	2.335847	1.307184
MA-5	2.245309	1.248511
WMA-4	2.428521	1.397725
WMA-5	2.2891	1.314257
EA	2.426604	1.339139

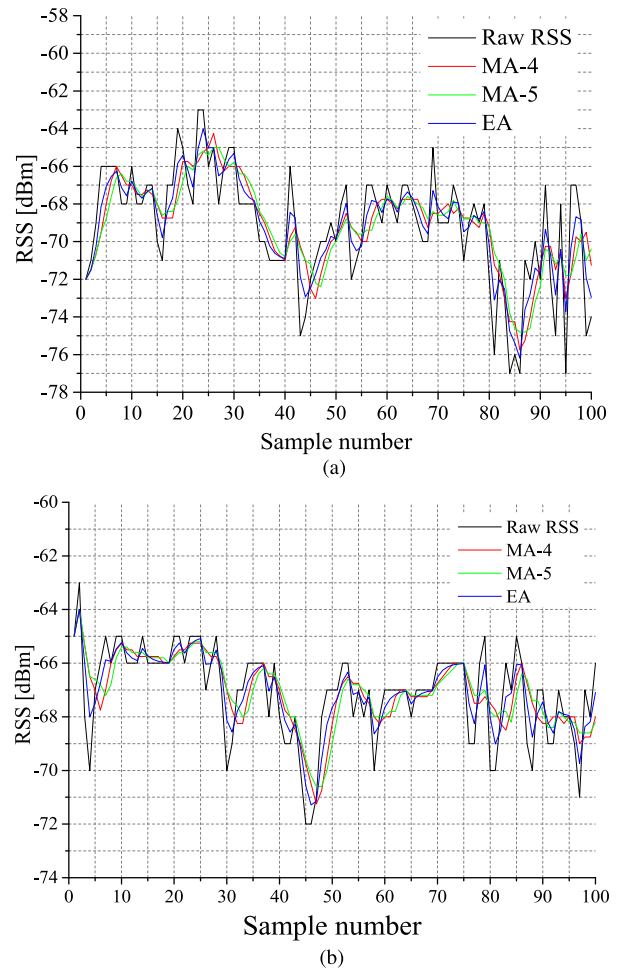
Fig. 5 shows the graphical representation of the observed RSS with various filters. For clarity, we have omitted the result of WMA in the graph.

It can be observed in Fig. 5 that the smoothing operation has reduced the spikes of raw RSS. It can also be observed that the result of EA is comparable to the results of MA and WMA with window sizes of 4 and 5. In addition, the smoothing operation in EA smoothing can yield results with only the first two RSS observations, whereas MA and WMA hold for window-sized RSS observations. Therefore, we used the EA filter for RSS smoothing for further experiments in this study.

B. EXPERIMENTAL RESULTS

This section presents the experimental results obtained by implementing the proposed positioning method. Fig. 6 shows the graphical representation of the testbed which consists of the long hallway/corridor and a computer lab. The center of the hallway consists of elevators, stairs, vending machines, and lavatories. It can be inferred that the people movement at the center of hallways is more frequent compared to another part of the corridor. Hence, we categorized our testbed as three cases; the fully furnished computer lab, the center of the hallway, and the remaining part of the hallway where each case have different RF attenuation environment. In Fig. 6, red, black, and green boundaries depict the computer lab, hallway/corridor, and the center of the hallway, respectively.

The parameters defined in (10) such as A and n were calibrated using received data in the offline phase. For A , RSS was recorded in all four directions of a beacon at one meter

**FIGURE 5.** The plot of smoothed RSS from various filters in different testbeds. (a) Computer lab. (b) Corridor.

distance and average result was calibrated. n was measured at various distances from the beacon (1 to 25 meters) and linear regression was used to calibrate the final value. The calibrated values of A and n were -62.91 dBm and 1.45 , respectively. Note that, since same set of calibrated values are reflected in the online positioning phase too, offline data were recorded from the computer lab and the corridor for the calibration of A and n , and the same set of calibrated values were used in whole testbed. In other words, the parameters of the propagation model were calibrated with representative data from all testbed environments and used uniformly in all the testbed environments owing to the fact that the same set of parameters are employed to compare offline and online WC. Depending on the beacons deployment, the value of g in (1) was assumed as 0.7 for offline and online phases. In addition, considering the deployment of beacons in rectangular fashion, the total number of nearby beacons u was set to four.

1) SIZE OF DATABASE WITH PROPOSED FINGERPRINTING POSITIONING

In a typical fingerprinting positioning, a small default value (usually less than -90 dBm) is assigned to the undetected

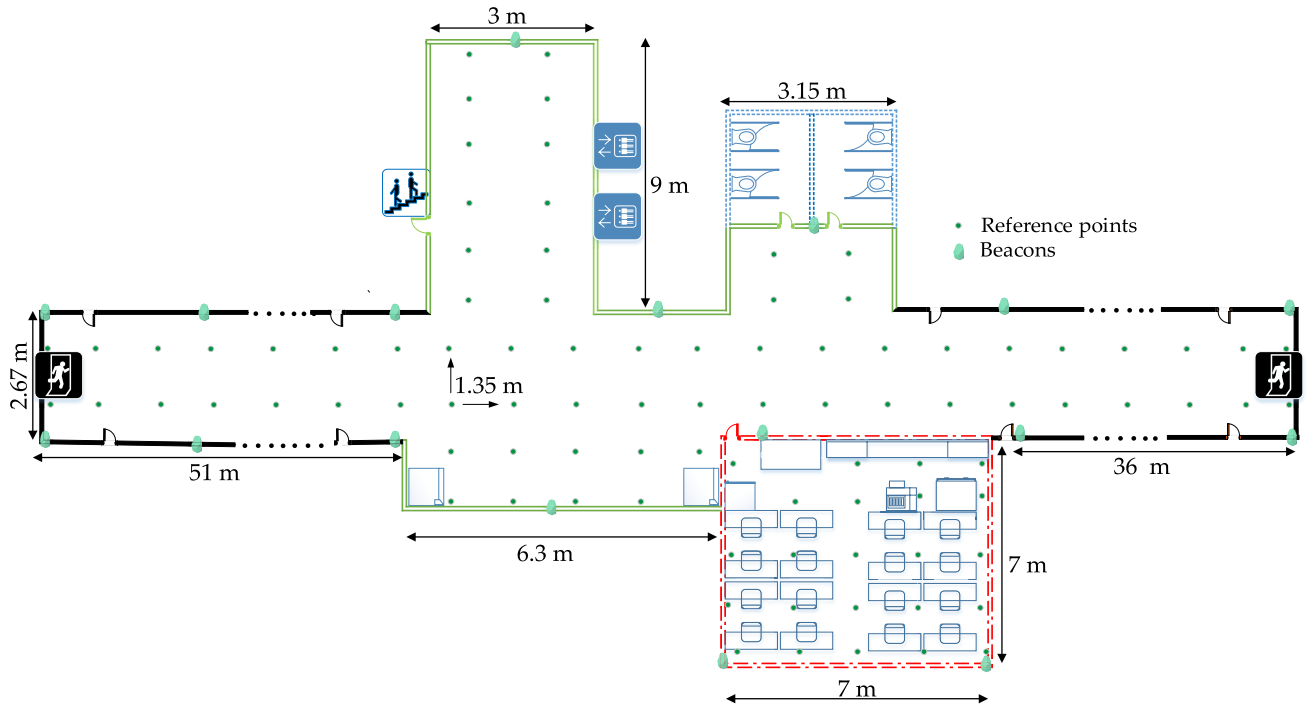


FIGURE 6. Graphical representation of the testbed.

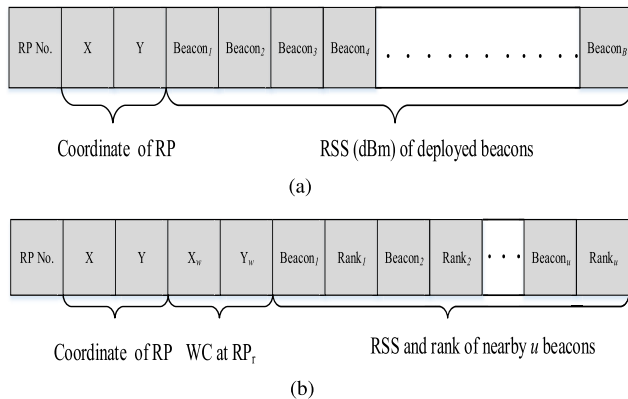


FIGURE 7. Columns of tables inside the fingerprinting database. (a) Typical RSS-based fingerprinting. (b) Proposed fingerprinting method.

APs, which are called invalid data [31]. Such invalid data is completely omitted in the database of the proposed method, which helps to reduce the size of the database. Moreover, as compared to the conventional fingerprinting localization, the signal strength information from $B - u$ (B being total deployed beacons) beacons at the RPs are redundant in the proposed method. We used Oracle Database 10g Express to store the fingerprint data in our system [42]. The columns of the tables inside the database were organized as shown in Fig. 7.

As shown in Fig. 7, each row in the database table contains the fingerprinting data of an RP. Note that each cell in the database tables is specified using numeric datatype. Moreover, we obtained the size of the physical memory

occupied by the fingerprinting database using SQL query as presented in Table 3.

TABLE 3. Physical memory occupied by fingerprinting database.

Parameter	Typical method	RSS-based	Proposed method
Avg. size of an RP	158.9 bytes		48.2 bytes
Size of fingerprinting database	32,413 bytes		9,830 bytes

As presented in Table 3, the physical size of the fingerprinting database of the proposed method ($u = 4$) can be reduced by 69.67% compared to the typical fingerprinting method ($B = 37$) under our experimental environment.

2) PERFORMANCE OF PROPOSED FINGERPRINTING POSITIONING

We evaluate and compare the proposed method in terms of positioning time (processing time consumed for positioning) and localization estimation error. For the positioning time calculation, we declared an *object* of *NSDate* [43] called *positioningStartTime* at the beginning, and the next *object* called *positioningFinishTime* at end of the positioning method in our positioning application. Later, we use *NSTimeInterval* to calculate the positioning time as the elapsed time between *positioningStartTime* and *positioningFinishTime* in seconds. The positioning time of the proposed method compared to some well-known existing methods, where the positioning time of the proposed method is t seconds is presented in Fig.8.

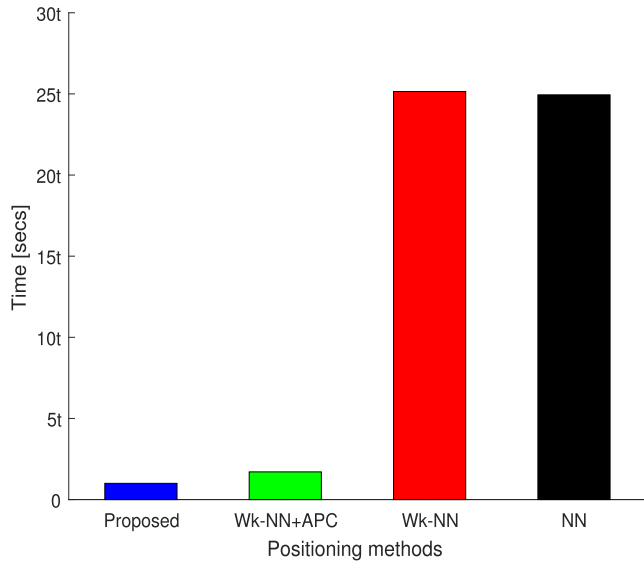


FIGURE 8. Performance comparison between the positioning methods in terms of positioning time.

As seen in Fig.8, the positioning time of the proposed method is lesser than the time taken by Wk-NN with APC (Wk-NN+APC). It shows that the proposed method is computationally effective compared to Wk-NN+ACP. Moreover, the positioning time of Wk-NN and nearest neighbor (NN) fingerprinting is very large compared to Wk-NN+APC and the proposed method. We noted that the positioning time incurred by Wk-NN is slightly larger than NN.

Various values of k in Wk-NN have been suggested such as $k = \{2\}$ and $k = \{3, 4\}$ [15], [44]. In our testbed, we achieved the least localization estimation error with $k = 3$. Thus, k was set as 3 in all experiments. We evaluated the performance of the proposed method in terms of localization error by considering three cases in our testbed as follows:

- 1) Hallway/corridor : Two walls bound the corridor and each wall contains metallic doors at 6.5 m distance intervals. Fig. 9 presents the cumulative probability and average location estimation error of the proposed method compared with existing methods. The NN fingerprinting yielded the location of an RP that has close resemblance with the online observed RSS data. In Wk-NN, k RPs that have least distance with the online RSS are selected and the final location is estimated by the weighted average of the k RP locations. Note that Wk-NN with APC was also implemented for the better comparison of the proposed method.

As seen in Fig. 9, the proposed method has a probability of 63% that localization estimation error is below 1m. Similarly, the Wk-NN with and without APC have probability of 55% and 22.5% for the same 1m positioning error condition, respectively. This result clearly shows the effectiveness of RSS clustering in indoor positioning. Moreover, average localization estimation error of the proposed method, positioning with

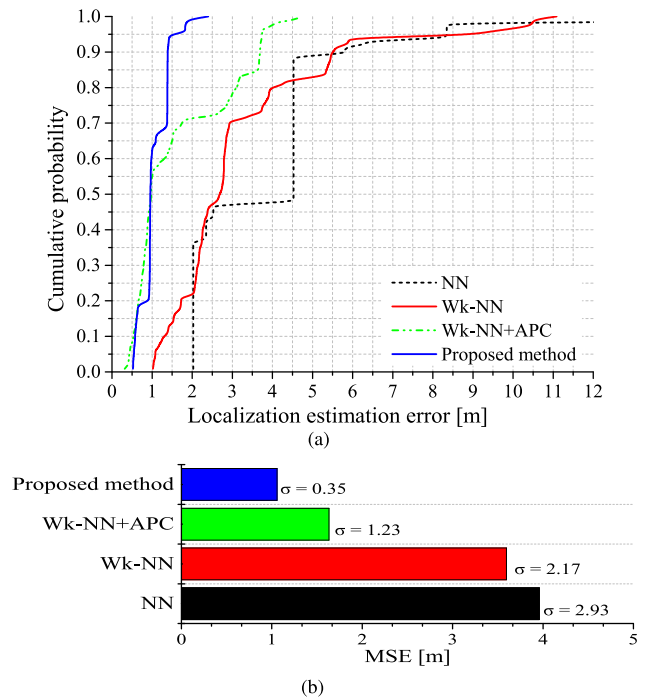


FIGURE 9. Localization estimation result at the hallways in the testbed. (a) Cumulative probability. (b) Average localization estimation error.

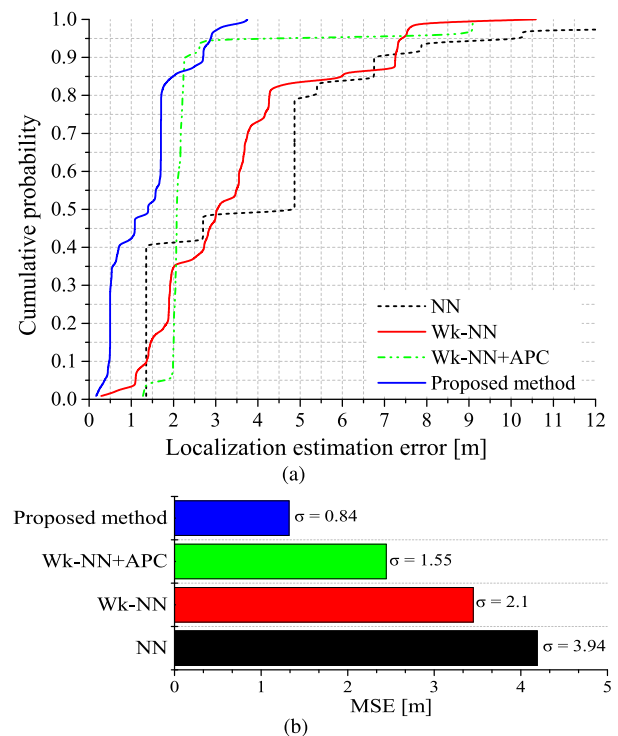


FIGURE 10. Localization estimation result at the center of hallways in the testbed. (a) Cumulative probability. (b) Average localization estimation error.

Wk-NN+APC, Wk-NN, and NN are 1.06m, 1.63m, 3.59m, and 4m, respectively.

- 2) Center of hallway/corridor: We choose to evaluate the proposed method at the center of hallway due to

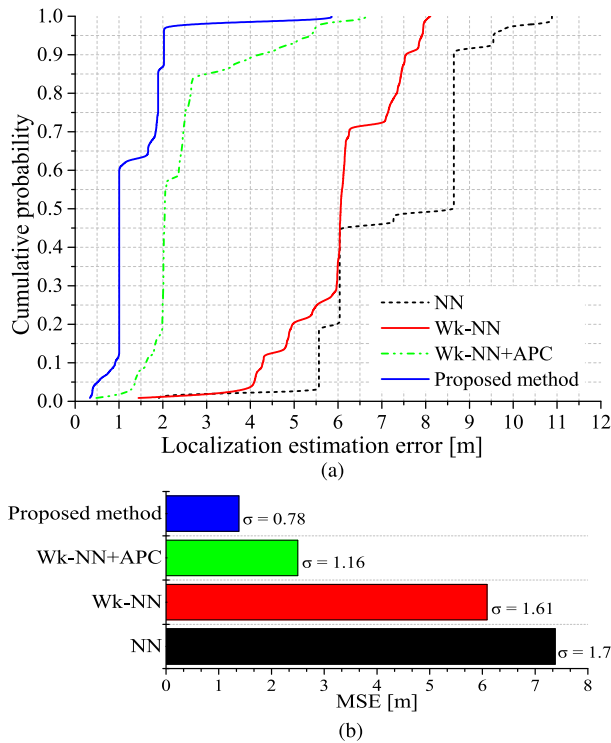


FIGURE 11. Localization estimation result at the computer lab in the testbed. (a) Cumulative probability. (b) Average localization estimation error.

presence of different public amenities that may result in different RF attenuation compared to another part of the hallway. Also, the center of corridor (marked by green boundry in Fig. 6) is more wider and crowded. As expected, the average localization estimation error has been slightly increased as compared to the other parts of the corridor as shown in Fig. 10. In particular, the proposed method and Wk-NN+APC have 1.32 m and 2.46 m of average localization estimation error, respectively.

- 3) Computer lab: The computer lab is the third case of our experiment. Since it is well furnished, more RF attenuation is expected compared to the corridor. The cumulative probability of the positioning methods as seen in Fig. 11 clearly depicts advantage of using APC for location estimation. The localization accuracy of Wk-NN and NN were significantly deteriorated where as Wk-NN+APC and the proposed method yielded acceptable result. It is noteworthy that the proposed method has probability of 2.5% that localization error is above 2.5 m. It can be due to the RF attenuation in the computer lab and wrong choice of cluster-head on online positioning phase.

Note that σ in Fig. 9, 10, and 11 refer to standard deviation of localization estimation error. In all the three cases, the positioning estimation of the proposed method is better than the methods under comparison. Particularly, the proposed method outperforms Wk-NN+APC method due to the following benefits:

- Diverse fingerprinting features: The proposed method employs WC, RSS, and rank of nearby beacons to define the fingerprint of an RP. Those fingerprinting informations was wisely used in (15), (16), and (17) to yield better positioning result.
- Better selection of cluster-head on online phase: As elaborated in Section III (A), the WC is confined inside the beacon locations. Hence, at any unknown place r' on positioning phase, it is likely to select a correct cluster-head with use of online-observed WC.

VI. CONCLUSION

In this paper, we presented the use of WC, RSS, and rank of nearby or locally available/seen beacons as a feature that defines RPs in fingerprinting positioning in indoor locations. The proposed method ensures that the fingerprint database is free from RSS data from distant beacons and reduces the size of the database. Our method employs APC to minimize the online computational cost of the system. The diversity on fingerprinting information along with RSS clustering in the proposed method helped to yield average localization estimation error of 1.05 m in the corridor and 1.38 m in the fully furnished computer lab.

We used BLE technology in our system owing to its low energy consumption and low-cost benefits. The localization result shows that the performance of the proposed method was better than Wk-NN+APC method.

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